



Rapport-Building Dialogue Strategies for Deeper Connection: Integrating Proactive Behavior, Personalization, and Aizuchi Backchannels

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Abstract

Proactive behavior, dialogue personalization, and backchannels are essential elements for enhancing rapport, fostering engagement, preventing conversation stalls, and supporting a natural dialogue flow. This study integrates these behaviors into a rapport-building dialogue strategy using the CO-STAR and few-shot frameworks to prompt large language models (LLMs) within a human-robot interaction. The success of the integration was assessed through metrics such as conversation stall frequency, dialogue similarity, and backchannel usage by the robot. We then observed the effect of integrating these elements on participant-observed behavioral outcomes, including participant backchannel usage, self-disclosure, and subjective questionnaire responses. Experimental results indicated that incorporating these behaviors into the dialogue strategy positively impacts human-robot rapport both behaviorally and subjectively.

Index Terms: Backchannels, personalization, proactive behavior, rapport-building dialogue strategy

1. Introduction

Rapport is defined as a state of mutual understanding, trust, and connection between individuals during an interaction [1]. Building rapport through shared understanding, trust, and engagement is essential for effective human-agent collaboration [2]. Baihaqi et al. introduced an early rapport-building dialogue strategy (RBDS) that employs specific rapport-promoting utterances—such as praise, self-disclosure, and storytelling [3]. This strategy is implemented within large language models (LLMs) through a free-form dialogue approach, designed to facilitate rapport during first meeting small talk.

However, without a specific strategy, the agent still faces challenges in sustaining longer conversations with passive participants. To prevent conversation stalls and effectively build rapport, it is important to personalize the dialogue by proactively collecting and utilizing the interlocutor's preferences while maintaining their engagement to construct responses [4]. Incorporating backchanneling, such as aizuchi (interjections indicating active listening in Japanese conversation), can further enhance interaction by fostering engaging conversation [5].

In this study, we sought to improve the RBDS by integrating proactive behavior, personalization, and aizuchi. This enhancement can be achieved by restructuring the prompt. By revamping the prompt structure with CO-STAR [6, 7], the agent gains a clear understanding of context, objectives, and tone, allowing it to take initiative in conversation and respond dynamically to user cues. This structure empowers the agent to personalize dialogue by adapting its style and responses to the user's interests, creating a more engaging and tailored interaction. In the few-

shot examples, we explicitly included aizuchi usage to help the LLM learn how to incorporate these interjections.

We integrated this dialogue system into a conversational robot, CommU, and compared the proposed strategy with various LLMs, against the baselines. Finally, we evaluated the impact of these behavioral modifications on human subjective responses and behavioral outcomes to assess improvements in rapport-building.

2. Related works

2.1. Proactive behavior and dialogue personalization

Proactive behavior is a crucial factor in enhancing interactions and preventing negative outcomes [8]. In provider-customer relationships, it is essential to build rapport and avoid service failure [9]. Similarly, in coach-athlete relationships, proactive behavior supports positive rapport and engagement [10]. In dialogue, proactive behavior involves introducing new topics through questions or statements to prevent dialogue stalls.

When making questions and statements, Humans are naturally adjusted based on the interlocutor, making interactions personalized and unique [11]. In investigative interviews, personalized conversations have been shown to foster higher rapport and increase self-disclosure [4]. Considering these effects, we integrate proactive behavior and personalized dialogue into the system's responses to prevent conversational stalls and ensure alignment with the user's interests.

2.2. Aizuchi in Japanese conversation

In Japanese conversation, aizuchi refers to a short verbal response that listeners use to signal attentiveness, understanding, and emotional responsiveness [12, 13, 14]. Aizuchi fosters a sense of unity and alignment within the conversation, helping to establish rapport and maintain relational harmony [5]. When aizuchi is absent, it may give the impression of disinterest, potentially weakening the rapport [12].

Aizuchi can be classified into several types [12]. Non-lexical (NL) aizuchi, like “うん” (un - yes) or “ええ” (ee - yes), uses to indicate attentiveness. Short responses (SR), such as “そうですね” (sou desu ne - that is right), signal understanding or agreement, while repetition (R) aizuchi involves echoing keywords to affirm the speaker's message. Combination aizuchi blends these forms, as in “うん、そうですね” (un, sou desu ne - yes, that is right), to convey deeper engagement.

Previous research has incorporated aizuchi in dialogue systems [15, 16], primarily focusing on non-lexical aizuchi and short responses. In contrast, this study designs a dialogue system that integrates various aizuchi types and investigates their impact on human responses.

3. Proposed methods

3.1. Integrating proactivity, personalization, and aizuchi into RBDS

The rapport-building dialogue strategy incorporated rapport-fostering utterances from human interactions, such as praise, storytelling, and jokes [3]. These elements were integrated into a free-form dialogue approach by prompting the LLM before the small talk, enabling the LLM to generate responses aligned with the initial prompt throughout the dialogue.

The original RBDS instructed the LLM to use both open-ended and short questions to maintain dialogue continuity. However, this approach was ineffective, as the conversation faltered when participants were passive. Furthermore, although the LLM attempted to engage users using 5W+1H questions to gauge user interests, it did not achieve the intended level of personalization. This was possibly attributed to the prompt's complexity and excessive commands, which made it difficult for the LLM to discern the focal point, leading to a loss of context [17].

To elicit proactive behavior and personalize dialogue, we revamped the prompt using the CO-STAR framework, which incorporates Context, Objective, Style, Tone, Audience, and Response components [6, 7]. Instead of adding additional commands that could complicate the prompt and reduce its clarity, this structured approach provides a clearer focal point for the LLM [18]. It also ensures that all key aspects are considered when generating the LLM's response, effectively aligning the output with the goal [19, 20].

The prompt structure is as follows:

- **Context:** The context is set to engage users in small talk about a specific dream travel destination, aiming to create a positive and engaging conversation.
- **Objective:** The primary goal is to build rapport by delivering a light, informal, and personalized conversation in Japanese, ensuring it aligns with the user's preferences and interests.
- **Style:** The conversation style is casual, friendly, and natural, resembling the tone of a chatty travel blogger, which helps make the dialogue relatable and approachable.
- **Tone:** The tone is light-hearted, enthusiastic, and empathetic to ensure a relaxed atmosphere for the user.
- **Audience:** The target audience consists of young adults who are interested in travel, preferring informal, inspiring, and engaging conversations about the destinations.
- **Response:** The responses should integrate rapport-building strategies and sustain engagement through follow-up questions and personalized details.

We also aimed to incorporate aizuchi backchannels into the responses. To achieve this, we employed few-shot learning by providing the LLM with two Japanese conversations that effectively utilized aizuchi. This approach enabled the model to better integrate these conversational cues.

3.2. Multimodal conversational robots

We embedded our dialogue systems into a robot developed by Vstone, CommU [21]. The robot was equipped with automatic speech recognition (ASR), text-to-speech (TTS), and predefined gesture modules, enabling it to function as a dialogue robot.

3.2.1. Robot types

This study evaluates four dialogue robots designed to build rapport in small talk over 28 turns discussing travel destinations.

The robots are as follows:

- **CSFS4o:** Uses the CO-STAR and few-shot framework with GPT-4o to elicit RBDS, focusing on proactive engagement, personalization, and increased use of aizuchi backchannels.
- **CSFS35:** Implements the same frameworks as CSFS4o but with GPT-3.5 Turbo. Comparing CSFS4o and CSFS35 allows us to evaluate the impact of model differences.
- **ORG35:** Applies the original RBDS [3] with GPT-3.5 Turbo, without specific strategies for proactive engagement, personalization, or aizuchi backchannels, allowing us to evaluate whether the addition of these behaviors enhances rapport.
- **ORG4o:** Applies the original RBDS with GPT-4o, enabling the assessment of whether improvements in dialogue quality stem from the LLM's enhanced capabilities or the inclusion of proactive engagement, personalization, and backchannels.

GPT-4o offers advantages in context management and prompt comprehension, enabling it to handle longer contexts and interpret longer prompts more effectively [22]. It is also perceived as more human-like, making its responses seem natural and expressive [23, 24]. Its broader knowledge base also enhances its ability to generate creative and engaging responses.

In contrast, GPT-3.5 Turbo is a smaller model with less ability to handle complex prompts. It was found to be semantically less rich, less emotional, and had less knowledge compared to its successor [25, 26, 27], resulting in less engaging responses.

3.3. Metrics evaluation

To demonstrate the successful integration of proactive behavior, personalization, and aizuchi, we used the following metrics:

1. **Conversation stall:** Conversation stall occurs when participants begin to provide brief responses, and the robot starts repeating similar responses due to a lack of new topics. Proactive robot behavior is expected to mitigate these stalls.
2. **Dialogue similarities:** Personalized dialogues are expected to have lower similarity due to user-specific responses. We measured the similarity by vectorizing each conversation and calculating the cosine similarity between dialogues.
3. **Aizuchi usage of robot:** We analyzed the frequency and type of aizuchi used by the robot.

We also analyzed the impact of integrating proactive behavior, personalization, and aizuchi on participant engagement using the following metrics:

1. **Aizuchi usage of participant:** The frequency and types of aizuchi used by participants were also analyzed.
2. **Effect on self-disclosure:** Given that the experiment was conducted in Japanese, self-disclosure was quantified by the total characters participants uttered per dialogue.
3. **Questionnaires:** We administered pairwise rapport and user experience questionnaires developed by Baihaqi et al. [3, 28]. Three rapport items were added to evaluate perceived proactivity and personalization: "I felt the robot was proactive in maintaining the conversation" (R8), "I felt the robot personalized the conversation for me" (R9), and "The robot appropriately introduced new topics as needed" (R10).

4. Experimental settings

4.1. Participants

We recruited 20 participants (10 males and 10 females, average age = 23.35, $SD = 1.02$) for our dialogue experiments. None had

Table 1: Frequency of conversation stall

Robot	Stall Frequency (times)	Average Turn to Stall
CSFS4o	0	-
CSFS35	4	21
ORG4o	7	20.29
ORG35	4	15

Table 2: Aizuchi usage by robot

	NL	SR	R	SR+R	SR+NL	NL+R
CSFS4o	23	105	10	14	24	15
CSFS35	1	57	23	30	1	0
ORG4o	3	54	8	13	1	1
ORG35	0	54	6	15	0	0

Table 3: Aizuchi usage by participants

	NL	SR	R	SR+R	SR+NL	NL+R
CSFS4o	11	99	7	2	11	0
CSFS35	8	95	3	1	2	0
ORG4o	5	78	1	3	4	0
ORG35	3	69	0	1	8	0

prior experience interacting with robots. Participants provided consent for the use of their data prior to the experiment.

4.2. Experimental procedure

We employed a counterbalanced design, where each participant interacted with each robot in a randomized order to minimize potential order bias. After completing interactions with all four robots, participants filled out the questionnaires. This experiment was conducted from October 14 to 24, 2024.

5. Experimental results

5.1. Robot proactivity, personalization, and aizuchi usage

5.1.1. Frequency of conversation stall

The frequency of stalls for each robot is shown in Table 1. CSFS4o had no conversation stalls. CSFS35 experienced four stalls, with an average stall occurring at turn 20.29. ORG4o had the most stalls, with seven, while ORG35 had only four stalls. However, ORG35 showed the earliest average stall, at turn 15.

5.1.2. Dialogue personalization

The average dialogue similarities across the robots were as follows: CSFS4o (7.48%), CSFS35 (8.26%), ORG4o (11.97%), and ORG35 (11.83%). The results indicate that robots utilizing CO-STAR and few-shot learning (CSFS4o and CSFS35) exhibited lower dialogue similarities compared to ORG4o and ORG35. Additionally, when examining unique words, CSFS4o featured 1,548 unique words, while CSFS35 had only 1,105. This suggests that CSFS4o generates more context-specific dialogues, whereas CSFS35 relies more on general responses.

5.1.3. Usage of aizuchi

Table 3 presents the distribution of aizuchi types used by each robot. All robots mainly utilized the SR type of aizuchi. However, CSFS4o exhibited greater variety in aizuchi usage, partic-

ularly in SR+NL and NL+R combinations, which were rarely used by the other robots. Overall, CSFS4o demonstrated the highest total aizuchi usage (188), followed by CSFS35 (123), ORG4o (87), and ORG35, which had the lowest usage (81).

5.2. Effect on experiment participants

5.2.1. Effect on participants' aizuchi

An interesting phenomenon was observed in the usage of aizuchi by the participants. As shown in Table 3, participants tended to use aizuchi more frequently when interacting with robots that also employed aizuchi more. The highest number of aizuchi was observed during interactions with CSFS4o (130 instances), followed by CSFS35 (109), ORG4o (87), and ORG35 (80). Additionally, consistent patterns in aizuchi usage were identified, with participants predominantly using SR-type aizuchi. This observation highlights how an increased usage of aizuchi by robots may influence participant behavior.

5.2.2. Effect on participants' self-disclosure

We quantified self-disclosure with the total uttered characters. On average, participants uttered the most characters during interactions with CSFS4o (412.8), followed by CSFS35 (391.4), and the least with ORG35 and ORG4o (386.1 and 359.9).

In Japanese conversation, even small differences, such as the 52.9-character gap between CSFS4o and ORG4o, can carry significant linguistic implications. For instance, *行きたい* (ik-itai) simply expresses "I want to go," while the addition of just one character in *行きたいよ* (ikitai yo) adds emphasis, making it sound more expressive, as in "I really want to go!" [29].

5.2.3. Questionnaire results

To analyze the UX and rapport questionnaire results presented in Figures 1 and 2, we conducted a repeated-measures ANOVA. Significant differences were observed across all aspects ($p < 0.05$ or $p < 0.01$), primarily attributed to ORG35's lower performance. For the overall UX scores, calculated as the mean of N1-U7, the results were as follows: CSFS4o = 3.43, ORG4o = 2.51, CSFS35 = 2.50, and ORG35 = 1.47. The analysis revealed a significant effect: $F(3, 57) = 21.8$, $p < 0.01$, and $\eta_p^2 = 0.53$. For the overall RS scores, calculated as the mean of R1-R10, the results were as follows: CSFS4o = 3.49, ORG4o = 2.44, CSFS35 = 2.53, and ORG35 = 1.44. The analysis revealed a significant effect: $F(3, 57) = 32.1$, $p < 0.01$, and $\eta_p^2 = 0.63$. These large effect sizes suggest that differences in robot behavior had a substantial impact on UX and RS.

We conducted head-to-head comparisons of each robot's questionnaire using a post-hoc Bonferroni test, as shown in Table 4. Overall, CSFS4o significantly outperformed the other robots, especially in naturalness, satisfaction, engagement, and certain rapport measures. Compared to CSFS35, CSFS4o demonstrated significant differences in UX ($p < 0.05$) and RS ($p < 0.01$). Against ORG4o, both UX and RS differences were significant ($p < 0.01$). Notably, compared to ORG35, CSFS4o was perceived as significantly better in all aspects ($p < 0.01$).

We did not find any significant differences between ORG4o and CSFS35 across all aspects. When comparing ORG4o with ORG35, $p < 0.05$ statistically significant differences were observed in satisfaction, dialogue interest, usability (U7), and several rapport questions (R1, R3, R4, R5, R6). A $p < 0.01$ difference was found in engagement, usability (U6), and four rapport questions, as well as overall UX and RS.

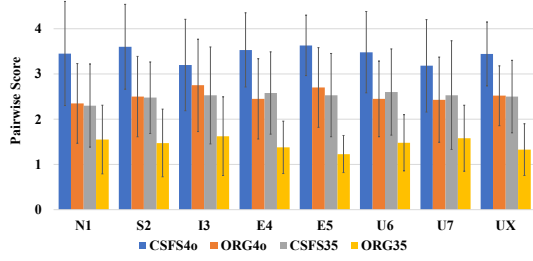


Figure 1: UX questionnaire results.

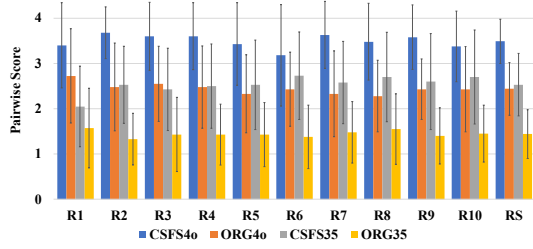


Figure 2: Rapport questionnaire results.

Table 4: Post-hoc Bonferroni test result

Aspect	AxO	AxI	AxE	IxO	IxE	OxE
N1	*	*	**	-	-	-
S2	*	*	**	-	*	**
I3	-	-	**	-	*	-
E4	-	**	**	-	**	**
E5	**	*	**	-	**	**
U6	-	*	**	-	**	**
U7	-	-	**	-	*	*
UX	*	**	**	-	**	**
R1	**	-	**	-	*	-
R2	**	**	**	-	**	**
R3	**	**	**	-	*	*
R4	*	**	**	-	*	**
R5	*	**	**	-	*	*
R6	-	-	**	-	**	**
R7	**	**	**	-	*	**
R8	-	**	**	-	-	*
R9	-	**	**	-	**	**
R10	-	*	**	-	**	**
RS	**	**	**	-	**	**

A: CSFS4o, O: CSFS35, I: ORG4o, and E: ORG35

*, $p < 0.05$, **, $p < 0.01$, -: no significant difference

Comparing CSFS35 with ORG35, we found significant differences at $p < 0.01$ in more aspects than with ORG4o and ORG35. Significant differences were observed in satisfaction, engagement, usability (U6), seven rapport questions, and overall UX and RS. $p < 0.05$ differences were found in usability (U7) and three rapport questions.

6. Discussion

Based on our experimental results, including the frequency of conversation stalls, dialogue similarities, and aizuchi usage by the robots, we infer that the elicitation of proactive behavior, personalization, and aizuchi backchannels was successful. Both proposed robots, CSFS4o and CSFS35, outperformed the base-

line (ORG35 and ORG4o), with CSFS4o—powered by GPT-4o—showing particular superiority, exhibiting zero stalls, the lowest similarity, a rich variety of words, and the highest and most varied usage of aizuchi.

Both CSFS4o and CSFS35 effectively used aizuchi and demonstrated proactive behavior by asking follow-up questions. However, CSFS4o’s responses were notably more creative and engaging. For example, when a participant mentioned an interest in traveling to the United States to watch baseball, CSFS4o not only acknowledged the statement but also asked a specific follow-up: “How about New York or Boston? You can enjoy watching the Yankees in New York and the Red Sox in Boston!” This question was tailored to the participant’s interest, mentioning specific teams and adding a personal touch¹.

In contrast, CSFS35 also used aizuchi and asked a follow-up question, but it was more general: “What other places would you like to go?” While this encouraged continued conversation, it lacked specificity. CSFS35’s open-ended question, focusing on any other possible venues, was less personalized, likely due to the limitations of the GPT-3.5 Turbo model. Although the response helped reduce dialogue similarity, it also resulted in a less deep interaction compared to CSFS4o.

Examining the effect on participants, both CSFS4o and CSFS35 garnered higher attention. Participants used aizuchi more frequently with CSFS4o, indicating greater engagement. Participants also showed less hesitation to express themselves when interacting with CSFS4o showed by longer responses.

Based on the questionnaire results, both UX and rapport scores revealed CSFS4o’s superiority over the other robots. The proactivity, personalization, and aizuchi significantly enhanced naturalness, engagement, usability, and ultimately, rapport. In interviews, participants noted the challenge in choosing them, as CSFS35 offered proactivity and aizuchi but lacked creativity, while ORG4o was more creative, albeit with less proactivity and aizuchi. Some participants noted that ORG4o exhibited a better user experience, while CSFS35 fostered rapport better. However, we found no statistical support for this. Finally, ORG35 performed the worst, with limited creativity, no proactivity, minimal personalization, and minimal use of aizuchi.

7. Conclusion

In this study, we introduced a rapport-building dialogue strategy that elicited proactivity, personalization, and aizuchi backchannels through CO-STAR and a few-shot framework. Our experimental results demonstrated the success of this approach, as evidenced by lower conversation stalls, reduced similarity, and an increased variety of aizuchi compared to the baseline. The results showed that proactivity, personalization, and aizuchi significantly influenced participants’ responses to the robots. Participants exhibited more self-disclosure, used aizuchi more frequently, and appeared more attentive, indicating enhanced rapport. Our questionnaire results revealed statistically significant improvements in user experience, with participants reporting more natural dialogues, greater satisfaction, higher engagement, and, ultimately, a higher perception of rapport.

Future research could enhance rapport and UX by exploring gesture strategies that complement RBDS. While this study used predefined gestures, they were limited and less natural. Future research in dynamic gestures that are contextually aligned with verbal expressions will be beneficial.

¹The dialogue sample is available at <https://github.com/yezato11/Rapport-Robot-II>

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